Retail Billing with Multiple Discount Rules

Complete worked solution

Sample rule: Electronics receives 12% category discount and 8% tax; clothing receives 5% discount and 5% tax; grocery receives no category discount and 2% tax. Apply an additional 5% basket discount when pre-tax subtotal reaches 10000, but total discount may not exceed 20% of pre-tax subtotal. Tax is calculated after discounts. Quantities must be positive and prices nonnegative.

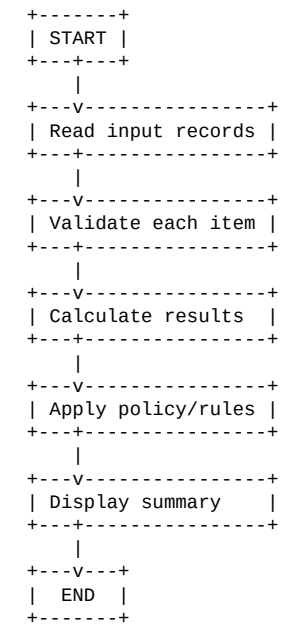
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read product tuples and discount cap
  FOR each product
    Validate quantity and price
    Find category discount and tax rate
    Calculate line subtotal and category discount
  Apply basket discount if subtotal reaches threshold
  Cap combined discounts at 20% of subtotal
  Allocate basket discount proportionally to product lines
  Calculate tax on discounted line amounts
  Print itemized invoice and total
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
# product: (name, category, unit_price, quantity)
products = [
    ("Headphones", "electronics", 4200, 2),
    ("Shirt", "clothing", 1800, 3),
```

```
    ("Rice bag", "grocery", 2500, 1),
]
subtotal = 0
category_discount_total = 0
lines = []
for name, category, price, quantity in products:
    if price < 0 or quantity <= 0:
        print("Invalid item:", name)
        continue
    line_subtotal = price * quantity
    discount_rate = 0
    tax_rate = 0
    if category == "electronics":
        discount_rate, tax_rate = 0.12, 0.08
    elif category == "clothing":
        discount_rate, tax_rate = 0.05, 0.05
    elif category == "grocery":
        discount_rate, tax_rate = 0, 0.02
    else:
        print("Unknown category:", category)
    category_discount = line_subtotal * discount_rate
    subtotal = subtotal + line_subtotal
    category_discount_total = category_discount_total + category_discount
    lines.append([name, category, quantity, price, line_subtotal,
                  category_discount, tax_rate])

basket_discount = subtotal * 0.05 if subtotal >= 10000 else 0
maximum_discount = subtotal * 0.20
allowed_discount = min(category_discount_total + basket_discount,
                       maximum_discount)
actual_basket_discount = allowed_discount - category_discount_total
if actual_basket_discount < 0:
    actual_basket_discount = 0
print("ITEM                 QTY  GROSS   DISCOUNT  TAX")
tax_total = 0
grand_total = 0
for line in lines:
    name, category, quantity, price, gross, category_discount, tax_rate = line
    # Share basket discount in proportion to each line's gross value.
    share = actual_basket_discount * gross / subtotal if subtotal > 0 else 0
    net_before_tax = gross - category_discount - share
    tax = net_before_tax * tax_rate
    total = net_before_tax + tax
    tax_total = tax_total + tax
    grand_total = grand_total + total
    print(name, quantity, round(gross, 2),
          round(category_discount + share, 2), round(tax, 2))
print("Subtotal:", round(subtotal, 2))
print("Total discount:", round(allowed_discount, 2))
print("Tax:", round(tax_total, 2), "AMOUNT DUE:", round(grand_total, 2))
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.